Series RLC Circuit

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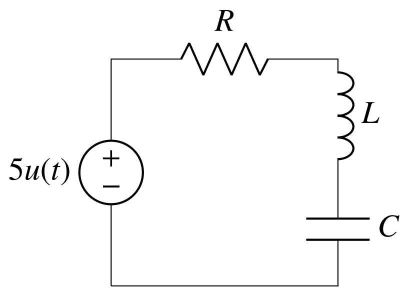
Abstract

This report explores the behavior of an RLC circuit exhibiting damped oscillations. The differential equation governing the circuit is derived, and solutions for different damping conditions are analyzed. Simulations and plots illustrate the circuit's response.

1 Introduction

An RLC circuit consists of an inductor (L) and a capacitor (C), forming an oscillatory system. When a resistor (R) is added, the circuit exhibits damped oscillations, leading to energy dissipation over time.

2 Theory for Underdamped Case



Consider a series RLC circuit connected to a unit step voltage source V_0. The switch is closed at $t = 0$, applying a step input $V_{in}(t) = V_0$ for $t \geq 0$.

By Kirchhoff's Voltage Law (KVL)

$$V_R(t) + V_L(t) + v_C(t) = V_0 \tag{1}$$

Using the fundamental current-voltage relationships for these components:

$$V_R = iR, \quad V_L = L\frac{di}{dt}, \quad i = C\frac{dv_C}{dt} \tag{2}$$

We can substitute the current i into the KVL equation. Since $i = C\frac{dv_C}{dt}$, the derivative of the current is $\frac{di}{dt} = C\frac{d^2v_C}{dt^2}$. Substituting these into equation (1) yields:

$$RC\frac{dv_C}{dt} + LC\frac{d^2v_C}{dt^2} + v_C = V_0 \tag{3}$$

$$\implies \frac{d^2v_C}{dt^2} + \frac{R}{L}\frac{dv_C}{dt} + \frac{1}{LC}v_C = \frac{V_0}{LC} \tag{4}$$

Undamped Natural Frequency:

$$\omega_n = \frac{1}{\sqrt{LC}} \text{ [rad/s]} \tag{5}$$

Damping Ratio (ζ) :

$$\zeta = \frac{R}{2L}\sqrt{LC} = \frac{R}{2}\sqrt{\frac{C}{L}} \tag{6}$$

Using ω_n and ζ, we can rewrite the differential equation (4) as:

$$\frac{d^2v_C}{dt^2} + 2\zeta\omega_n\frac{dv_C}{dt} + \omega_n^2 v_C = \omega_n^2 V_0 \tag{7}$$

General Solution

The general solution to this non-homogeneous differential equation is the sum of the steady-state (particular) solution v_{ss} and the transient (homogeneous) solution v_{tr}:

$$v_C(t) = v_{ss} + v_{tr} \tag{8}$$

Since the input is a constant DC step (V_0), the capacitor acts as an open circuit in the steady state ($t \to \infty$). Thus, the steady-state voltage is:

$$v_{ss} = V_0 \tag{9}$$

The transient solution v_{tr} is found by solving the characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \tag{10}$$

Using the quadratic formula, the roots $s_{1,2}$ are:

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \tag{11}$$

Note :

- For t<0 the input voltage is -2.5V not 0V because the offset kept is 0V not 2.5V

The nature of these roots and the resulting time-domain response depends entirely on the value of ζ. We now assume the capacitor is initially charged to the negative rail: $v_C(0) = -V_0$ and zero initial current $i(0) = C\frac{dv_C}{dt}(0) = 0$.

2.1 Overdamped ($\zeta > 1$)

When $\zeta > 1$, the roots s_1 and s_2 are real, negative, and distinct.

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \tag{12}$$

The total solution takes the form:

$$v_C(t) = V_0 + c_1e^{s_1t} + c_2e^{s_2t} \tag{13}$$

Applying Initial Conditions:
1. $v_C(0) = -V_0$:

$$-V_0 = V_0 + c_1e^0 + c_2e^0 \implies c_1 + c_2 = -2V_0 \tag{14}$$

2. $v'_C(0) = 0$:

$$v'_C(t) = s_1c_1e^{s_1t} + s_2c_2e^{s_2t} \implies s_1c_1 + s_2c_2 = 0 \implies c_2 = -c_1\frac{s_1}{s_2} \tag{15}$$

$$\implies c_1 - c_1\frac{s_1}{s_2} = -2V_0 \implies c_1\left(\frac{s_2 - s_1}{s_2}\right) = -2V_0 \implies c_1 = \frac{-2V_0s_2}{s_2 - s_1} \tag{16}$$

$$c_2 = -2V_0 - c_1 = -2V_0 - \left(\frac{-2V_0s_2}{s_2 - s_1}\right) = \frac{2V_0s_1}{s_2 - s_1} \tag{17}$$

Final Overdamped Solution:

$$\boxed{v_C(t) = V_0 - \frac{2V_0}{s_2 - s_1}\left(s_2e^{s_1t} - s_1e^{s_2t}\right)} \tag{18}$$

2.2 Critically Damped ($\zeta = 1$)

When $\zeta = 1$, the Discriminant becomes zero, resulting in two real, repeated roots:

$$s_1 = s_2 = -\omega_n \tag{19}$$

The total solution takes the form:

$$v_C(t) = V_0 + (c_1 + c_2t)e^{-\omega_nt} \tag{20}$$

Applying Initial Conditions:
1. $v_C(0) = -V_0$:

$$-V_0 = V_0 + (c_1 + 0)e^0 \implies c_1 = -2V_0 \tag{21}$$

2. $v'_C(0) = 0$:

$$v'_C(t) = c_2 e^{-\omega_n t} - \omega_n(c_1 + c_2 t)e^{-\omega_n t} \tag{22}$$

$$0 = c_2 - \omega_n c_1 \implies c_2 = \omega_n c_1 = -2V_0\omega_n \tag{23}$$

Final Critically Damped Solution:

$$\boxed{v_C(t) = V_0 - 2V_0(1 + \omega_n t)e^{-\omega_n t}} \tag{24}$$

2.3 Underdamped ($\zeta < 1$)

When $\zeta < 1$, the Discriminant becomes negative, yielding complex conjugate roots. We define the damped natural frequency ω_d:

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \tag{25}$$

The roots become $s_{1,2} = -\zeta\omega_n \pm j\omega_d$. The total solution is:

$$v_C(t) = V_0 + e^{-\zeta\omega_n t}(c_1 \cos(\omega_d t) + c_2 \sin(\omega_d t)) \tag{26}$$

Applying Initial Conditions:
1. $v_C(0) = -V_0$:

$$-V_0 = V_0 + e^0(c_1 \cos(0) + c_2 \sin(0)) \implies c_1 = -2V_0 \tag{27}$$

2. $v'_C(0) = 0$:

$$0 = -\zeta\omega_n c_1 + \omega_d c_2 \implies c_2 = c_1 \frac{\zeta\omega_n}{\omega_d} = -2V_0 \frac{\zeta\omega_n}{\omega_d} \tag{28}$$

Substitute $\omega_n/\omega_d = 1/\sqrt{1 - \zeta^2}$:

$$c_2 = -2V_0 \frac{\zeta}{\sqrt{1 - \zeta^2}} \tag{29}$$

Final Underdamped Solution:

$$\boxed{v_C(t) = V_0 \left[1 - 2e^{-\zeta\omega_n t}\left(\cos(\omega_d t) + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin(\omega_d t)\right)\right]} \tag{30}$$

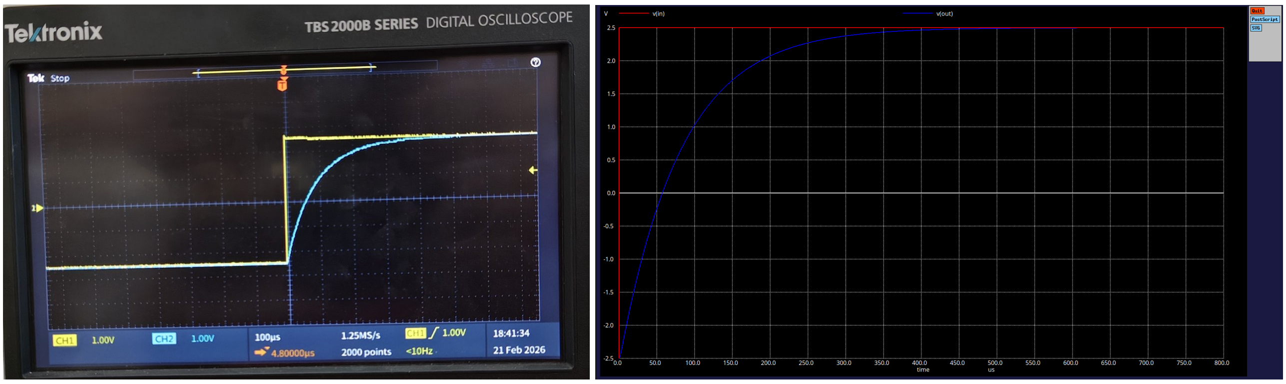
3 Experimental Waveforms

- According to values of capacitor and inductor available we have taken C = 10nF and L = 10mH
- The value of resistance taken depends upon the case we are analyzing

3.1 Overdamped ($\zeta > 1$)

- For this case we have taken R = 8.2k Ω
- $\zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$

$$\zeta = \frac{8200}{2}\sqrt{10^{-6}} = 4100 \times 10^{-3} = \mathbf{4.10} \tag{31}$$

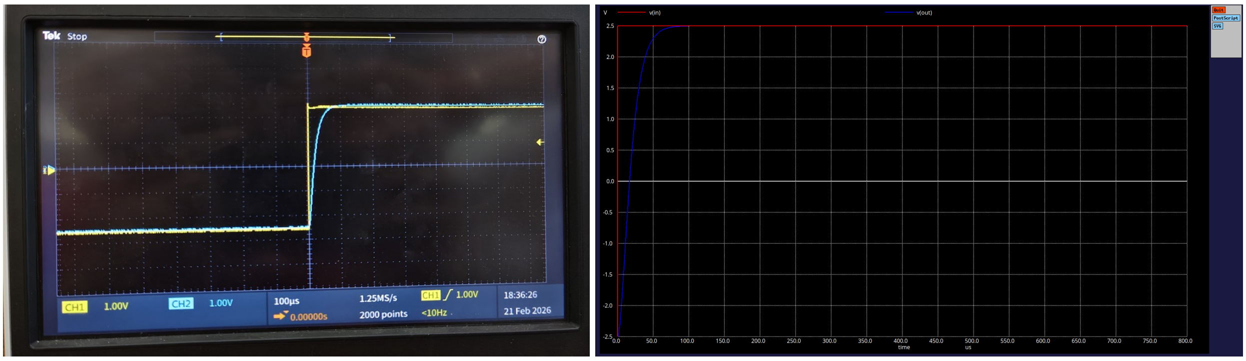


Experimental waveform and ngspice simulation

3.2 Critically damped ($\zeta = 1$)

- For this case we have taken R = 2k Ω

$$\zeta = \frac{2000}{2}\sqrt{10^{-6}} = 1000 \times 10^{-3} = \mathbf{1.00} \tag{32}$$

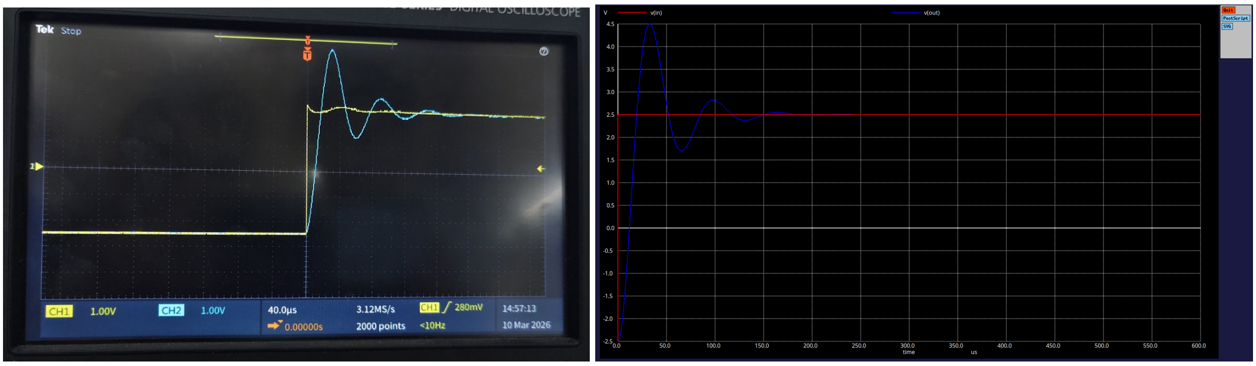


Experimental waveform and ngspice simulation

3.3 Underdamped ($\zeta < 1$)

- For this case we have taken R = 440 Ω

$$\zeta = \frac{440}{2}\sqrt{\frac{10\times10^{-9}}{10\times10^{-3}}} = 220\times10^{-3} = \mathbf{0.22} \tag{33}$$



Experimental waveform and ngspice simulation

3.4 Overshoot

- We will observe overshoot when the value of $\zeta < 1$
- We can find the peak overshoot by equating the value of $v_C(t)$ in this particular case to 0
- $v_C(t)$ for $\zeta < 1$ is given by

$$v_C(t) = V_0 - 2V_0 e^{-\zeta\omega_n t}\left(\cos(\omega_d t) + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin(\omega_d t)\right) \tag{34}$$

3.4.1 Derivation of Peak Time (t_p)

The peak occurs when the rate of change of the capacitor voltage is zero. We differentiate equation (34) with respect to time t:

$$\begin{aligned}\frac{dv_C}{dt} &= -2V_0\frac{d}{dt}\left[e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta\omega_n}{\omega_d}\sin\omega_d t\right)\right]\\ &= -2V_0 e^{-\zeta\omega_n t}\left[-\zeta\omega_n\cos\omega_d t - \frac{\zeta^2\omega_n^2}{\omega_d}\sin\omega_d t - \omega_d\sin\omega_d t + \zeta\omega_n\cos\omega_d t\right]\end{aligned}$$

use $\omega_d^2 = \omega_n^2(1-\zeta^2)$:

$$\frac{dv_C}{dt} = \frac{2V_0\omega_n^2}{\omega_d}e^{-\zeta\omega_n t}\sin(\omega_d t) \tag{35}$$

Setting $\frac{dv_C}{dt} = 0$, the first peak occurs when $\sin(\omega_d t) = 0$ for $t > 0$:

$$\omega_d t_p = \pi \implies \boxed{t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}} \tag{36}$$

3.4.2 Derivation of Maximum Peak Voltage (V_{peak})

To find the maximum voltage, we evaluate $v_C(t)$ at the peak time t_p. Substituting $t = t_p$ into equation (34), and noting that $\cos(\omega_d t_p) = \cos(\pi) = -1$ and $\sin(\omega_d t_p) = \sin(\pi) = 0$:

$$V_{peak} = v_C(t_p) = V_0 - 2V_0 e^{-\zeta\omega_n(\frac{\pi}{\omega_d})}(-1+0) \tag{37}$$

$$V_{peak} = V_0 + 2V_0 e^{-\left(\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right)} \tag{38}$$

Defining the total step height as $\Delta V = 2V_0 = 5V$, the peak voltage is:

$$\boxed{V_{peak} = V_{final} + \Delta V \cdot e^{-\left(\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right)}} \tag{39}$$

The percentage overshoot relative to the total step magnitude ΔV is:

$$\boxed{\%OS = e^{-\left(\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right)} \times 100\%} \tag{40}$$

3.4.3 Calculations :

Let′s calculate the peak overshoot of the circuit parameters used: $L = 10\,\text{mH}$, $C = 10\,\text{nF}$, and $R = 440\,\Omega$. The input steps from -2.5 V to $+2.5$ V, resulting in a total step height $\Delta V = 5$ V.

The damping ratio ζ is calculated as:

$$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}} = \frac{440}{2}\sqrt{\frac{10^{-8}}{10^{-2}}} = 220 \times 0.001 = \mathbf{0.22} \tag{41}$$

The percentage overshoot depends only on ζ:

$$\%OS = e^{-\left(\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right)} \times 100\% \tag{42}$$

Plugging in $\zeta = 0.22$:

$$\%OS = e^{-\left(\frac{0.22\times\pi}{\sqrt{1-0.22^2}}\right)} \times 100\% \approx e^{-\left(\frac{0.6912}{0.9755}\right)} \times 100\% \tag{43}$$

$$\%OS = e^{-0.7085} \times 100\% \approx 0.4924 \times 100\% = \mathbf{49.24}\% \tag{44}$$

The overshoot in volts is calculated relative to the total step height ($\Delta V = 5\,\mathrm{V}$):

$$V_{os} = \Delta V \times \frac{\%OS}{100} = 5\,\mathrm{V} \times 0.4924 = \mathbf{2.46}\,\mathrm{V} \tag{45}$$

The maximum voltage reached by the capacitor is the sum of the steady-state value ($V_{ss} = 2.5\,\mathrm{V}$) and the overshoot magnitude:

$$\boxed{V_{peak} = V_{ss} + V_{os} = 2.5\,\mathrm{V} + 2.46\,\mathrm{V} = 4.96\,\mathrm{V}} \tag{46}$$

3.4.4 Observed Results :

Measured peak overshoot with cursors

Error

$$\text{error } = \frac{2.5 - 2.46}{2.5} \times 100\% \tag{47}$$
$$= 1.6\% \tag{48}$$

3.4.5 Conclusion :

- So the maximum value of the capacitor voltage for our taken values is $\approx$ 5V theoretically which is almost equal(with an error < 2%) to the experimental value measured using cursors in the DSO

3.5 Presence of oscillations

- when $\zeta > 1$ the output will increase exponentially from -2.5V to 2.5V
- when $\zeta = 1$ the output will increase exponentially from -2.5V to 2.5V but has less settling time than the overdampped case
- when $\zeta < 1$ the output exhibits oscillations (ringing) with an exponentially decaying amplitude, eventually settling at the steady-state input voltage of 2.5V.

3.6 Settling Behaviour

The settling time t_s is defined as the time required for the response to reach and remain within a specified percentage (typically 2% or 5%) of its final steady-state value V_{ss}.
For a step transition from $-V_0$ to V_0(where $V_0 = 2.5$V), the total step height is:

$$\Delta V = 2V_0 \tag{49}$$

For a 2% settling criterion:

$$|v_C(t) - V_{ss}| \leq 0.02\,\Delta V \tag{50}$$

Thus,

$$|v_C(t) - V_0| \leq 0.04V_0 \tag{51}$$

3.6.1 Critically Damped ($\zeta = 1$)

For a critically damped system, the capacitor voltage is given by:

$$v_C(t) = V_0 - 2V_0(1 + \omega_n t)e^{-\omega_n t} \tag{52}$$

The transient term must satisfy the 2% error condition:

$$2V_0(1 + \omega_n t_s)e^{-\omega_n t_s} = 0.04V_0 \tag{53}$$

$$(1 + \omega_n t_s)e^{-\omega_n t_s} = 0.02 \tag{54}$$

The equation(52) is solved numerically. For a 2% settling criterion:

$$\omega_n t_s \approx 5.83 \tag{55}$$

Therefore,

$$\boxed{t_s \approx \frac{5.83}{\omega_n}} \tag{56}$$

If L = 10mH and C = 10nF then ,

$$\omega_n = \frac{1}{\sqrt{LC}} \tag{57}$$

$$\implies \omega_n = 10^5 \text{rad/s} \tag{58}$$

$$t_s \approx 58.3\,\mu s \tag{59}$$

Experimental value of settling time for $\zeta = 1 \approx 71\mu$s

Experimental settling time is almost equal to the settling time computed theoretically

3.6.2 Over Damped ($\zeta > 1$)

For a Over damped system, the capacitor voltage is given by:

$$v_C(t) = V_0 - \frac{2V_0}{s_2 - s_1}\left(s_2 e^{s_1 t} - s_1 e^{s_2 t}\right) \tag{60}$$

$\omega_n = 10^5$ rad/s and $\zeta = 4.1$:

$$s_{1,2} = \omega_n\left(-\zeta \pm \sqrt{\zeta^2 - 1}\right) \tag{61}$$

$$s_1 = 10^5\left(-4.1 + \sqrt{16.81 - 1}\right) \approx -12,348 \text{ rad/s} \tag{62}$$

$$s_2 = 10^5\left(-4.1 - \sqrt{16.81 - 1}\right) \approx -807,652 \text{ rad/s} \tag{63}$$

using (49)

$$0.02 \cdot \Delta V = \left|\frac{\Delta V}{s_2 - s_1}\left(s_2 e^{s_1 t_s} - s_1 e^{s_2 t_s}\right)\right| \tag{64}$$

$$0.02 = \left|1.0155 e^{s_1 t_s} - 0.0155 e^{s_2 t_s}\right| \tag{65}$$

Solving numerically:

$$t_s \approx \mathbf{318.5}\mu\text{s} \tag{66}$$

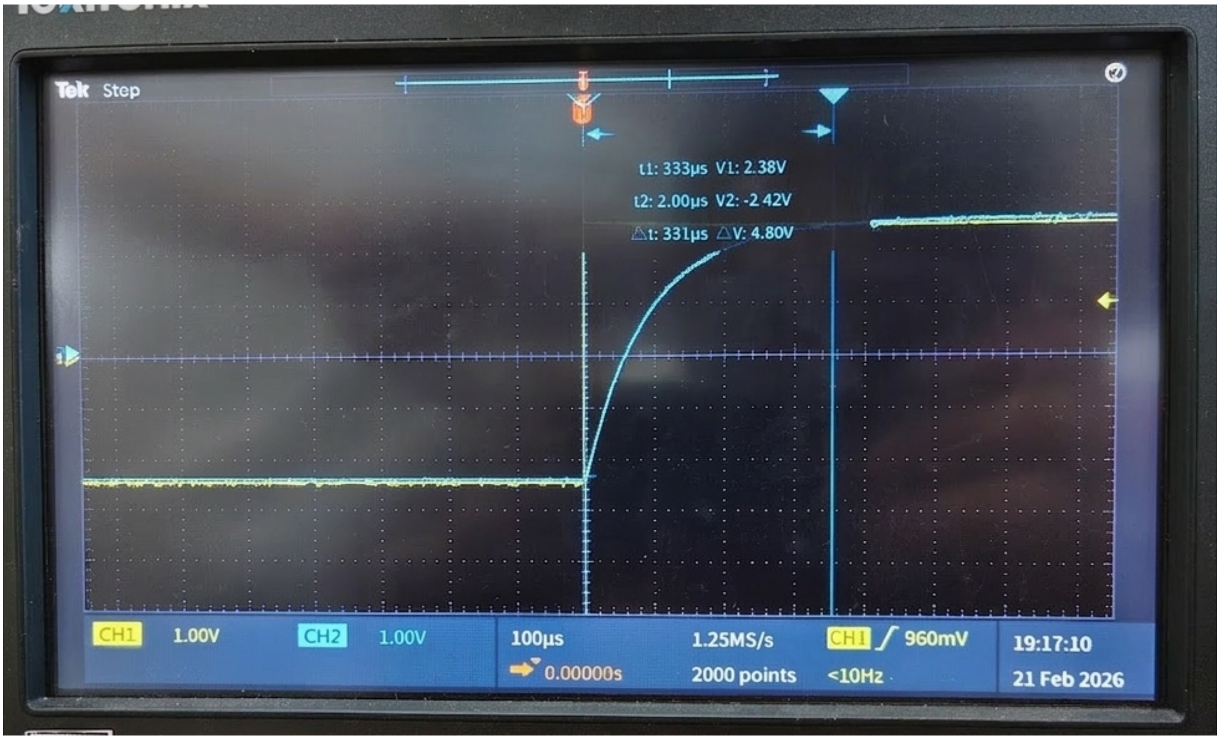


Figure 0: Experimental value of settling time for $\zeta > 1$

Experimental settling time is almost equal to the settling time computed theoretically

3.6.3 Conclusion

- The time taken by the output voltage to settle when $\zeta = 1$ is less than the time taken by the output voltage to settle when $\zeta > 1$

4 Frequency Response

- For the frequency response of the circuit in all the 3 cases we vary the frequency of the input wave from 10Hz to 10^5Hz and we plot the bodeplots for it

- we should also identify the bandwidth, Quality factor and resonant frequency

4.1 Quality Factor (Q)

The Quality Factor (Q) is fundamentally defined as 2π times the ratio of the maximum energy stored in the circuit to the energy dissipated per cycle at resonance:

$$Q = 2\pi \frac{W_{\text{stored}}}{W_{\text{dissipated per cycle}}} \tag{67}$$

4.1.1 Derivation

At resonance, the energy transfers continuously between the inductor and the capacitor. The total maximum stored energy W_{stored} can be evaluated when the current is at its peak I_m (at which point the capacitor's stored energy is zero and the inductor's energy is maximum):

$$W_{\text{stored}} = \frac{1}{2} L I_m^2 \tag{68}$$

The power dissipated by the resistor P_{avg} is:

$$P_{\text{avg}} = \frac{1}{2} I_m^2 R \tag{69}$$

$$W_{\text{dissipated per cycle}} = P_{\text{avg}} \cdot T_0 = \left(\frac{1}{2} I_m^2 R\right)\left(\frac{2\pi}{\omega_0}\right) \tag{70}$$

Substituting equations (68) and (70) into the definition of Q in equation (3):

$$Q = 2\pi \frac{\frac{1}{2} L I_m^2}{\left(\frac{1}{2} I_m^2 R\right)\left(\frac{2\pi}{\omega_0}\right)} \tag{71}$$

$$Q = \frac{\omega_0 L}{R} \tag{72}$$

Substituting $\omega_0 = \frac{1}{\sqrt{LC}}$

$$\boxed{Q = \frac{1}{R}\sqrt{\frac{L}{C}}} \tag{73}$$

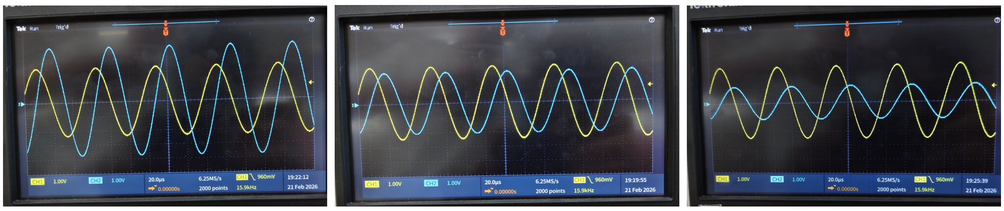
$$\boxed{Q = \frac{1}{2\zeta}} \tag{74}$$

4.1.2 Conclusions

- As the quality factor(Q) of the output curve increases the voltage graph gets more sharpened
- As the resistance decreases the quality factor increases and the peak value of the voltage will be increased
- A higher Quality Factor corresponds to a lower damping ratio (ζ), where $\zeta = \frac{1}{2Q}$. This relationship governs the transition of the system from a non-oscillatory overdamped state ($\zeta > 1$) to a highly oscillatory underdamped state ($\zeta < 1$) as resistance is reduced.

4.1.3 Experimental wave forms for Band Width

- The value of resistances taken for this purpose are 560Ω, 1000Ω, 2000Ω



Waveforms obtained for R = 560,1000,2000Ω

Verification

- The ratio of peak voltages of the output voltage in case 1 to case 2 is given by

$$\text{Ratio}_{Theo} = \frac{1000}{560} = 1.78 \tag{75}$$

- From the experimental waveforms

$$\text{Ratio}_{Exp} \approx \frac{3.7}{2.1} \approx 1.76 \tag{76}$$

- The ratio of peak voltages of the output voltage in case 2 to case 3 is given by

$$\text{Ratio}_{Theo} = \frac{2000}{1000} = 2 \tag{77}$$

- From the experimental waveforms

$$\text{Ratio}_{Exp} \approx \frac{2.1}{1.05} \approx 2 \tag{78}$$

- The experimental ratio ≈ Theoretical ratio which proves that the above calculations match the graphs obtained

4.2 Gain Peaking

Let us the Transfer function H(s) as follows

$$H(s) = \frac{V_o(s)}{V_i(s)} \tag{79}$$

The impedance of each element is

$$Z_R = R, \quad Z_L = sL, \quad Z_C = \frac{1}{sC}(\text{Where } s = \text{j}\omega\ )$$

$$H(s) = \frac{Z_C}{Z_R + Z_L + Z_C} \tag{80}$$

$$H(s) = \frac{1}{LCs^2 + RCs + 1} \tag{81}$$

$$\implies H(s) = \frac{1}{\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n}s + 1} \tag{82}$$

Comparing coefficients,

$$\omega_n = \frac{1}{\sqrt{LC}}$$

$$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$$

Substitute $s = j\omega$:

$$H(j\omega) = \frac{1}{1 - \frac{\omega^2}{\omega_n^2} + j\frac{2\zeta\omega}{\omega_n}} \tag{83}$$

The magnitude is

$$|H(j\omega)| = \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(\frac{2\zeta\omega}{\omega_n}\right)^2}} \tag{84}$$

Define the normalized frequency

$$x = \frac{\omega}{\omega_n} \tag{85}$$

Thus,

$$|H(j\omega)| = \frac{1}{\sqrt{(1 - x^2)^2 + (2\zeta x)^2}} \tag{86}$$

Gain peaking occurs when the magnitude response has a maximum at some finite frequency.

To find this maximum, minimize the denominator

$$D = (1 - x^2)^2 + (2\zeta x)^2 \tag{87}$$

Expanding,

$$D = 1 - 2x^2 + x^4 + 4\zeta^2 x^2 \tag{88}$$

$$D = 1 + x^4 + x^2(4\zeta^2 - 2) \tag{89}$$

Differentiate with respect to x:

$$\frac{dD}{dx} = 4x^3 + 2x(4\zeta^2 - 2) \tag{90}$$

$$\implies \frac{dD}{dx} = 2x(2x^2 + 4\zeta^2 - 2) \tag{91}$$

$$\implies 2x(2x^2 + 4\zeta^2 - 2) = 0 \tag{92}$$

$$\implies 2x^2 + 4\zeta^2 - 2 = 0 \tag{93}$$

$$\implies x^2 = 1 - 2\zeta^2 \tag{94}$$

For a real frequency peak,

$$x^2 > 0 \tag{95}$$

$$1 - 2\zeta^2 > 0 \tag{96}$$

$$2\zeta^2 < 1 \tag{97}$$

$$\boxed{\zeta < \frac{1}{\sqrt{2}}} \tag{98}$$

Thus, gain peaking occurs only when the damping ratio satisfies

$$\zeta < \frac{1}{\sqrt{2}}$$

The frequency at which the peak occurs is

$$\boxed{\omega_p = \omega_n \sqrt{1 - 2\zeta^2}} \tag{99}$$

The peak occurs at

$$x_p = \sqrt{1 - 2\zeta^2} \tag{100}$$

Substituting $x = x_p$ into the magnitude expression,

$$|H(j\omega_p)| = \frac{1}{\sqrt{(1 - x_p^2)^2 + (2\zeta x_p)^2}} \tag{101}$$

on simplyfing the above expression we get **gain peaking** (maximum gain) is

$$\boxed{|H(j\omega_p)| = \frac{1}{2\zeta \sqrt{1 - \zeta^2}}} \tag{102}$$

In decibels,

$$\text{Gain Peaking (dB)} = 20\log_{10}\left(\frac{1}{2\zeta\sqrt{1-\zeta^2}}\right) \tag{103}$$

Condition on ζ for gain peaking :

- $\zeta < 0.707$: System with gain peaking
- $\zeta \approx 0.707$: Maximally flat
- $\zeta > 0.707$: system with no peaking

4.3 Resonance Frequency

- Resonant frequency(ω_r) is defined as the frequency where gain peaking occurs if exist

$$\omega_r = \omega_n\sqrt{1-2\zeta^2} \tag{104}$$

- so Resonance condition exists only when the value of $\zeta < \frac{1}{\sqrt{2}}$

4.4 Band width

- The bandwidth (BW) is defined as the difference between the upper and lower half-power frequencies
- Band width exists only when the resonant peak exists because only in this case we can find the frequencies at which the power is half of the maximum power
- so if the value of $\zeta < \frac{1}{\sqrt{2}}$ then band width exists

At half-power frequencies, the current I is $1/\sqrt{2}$ of its maximum value $I_{max} = V/R$. This occurs when the magnitude of the impedance Z is $\sqrt{2}R$:

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2}R \tag{105}$$

simplifying, we get

$$BW_{\text{rad/s}} = \omega_2 - \omega_1 = \frac{R}{L} \tag{106}$$

$$\boxed{BW_{\text{Hz}} = \frac{R}{2\pi L}} \tag{107}$$

Note:
Though the above formula holds for over and critical damping also it is valid only mathematically physically it has no meaning as in the case of under damping

4.5 Critical Damping

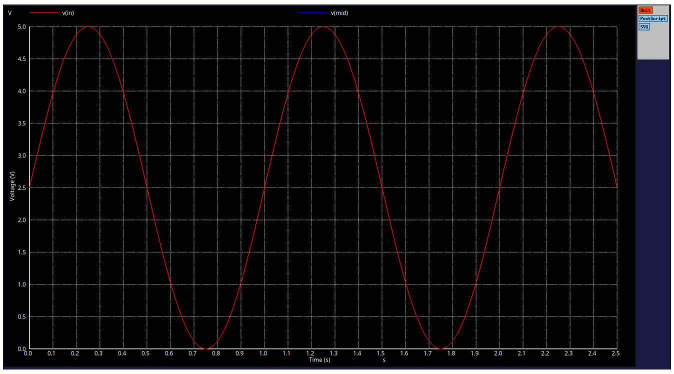
- For this case we have taken 2000Ω resistance

$$\zeta = \frac{2000}{2000} \tag{108}$$

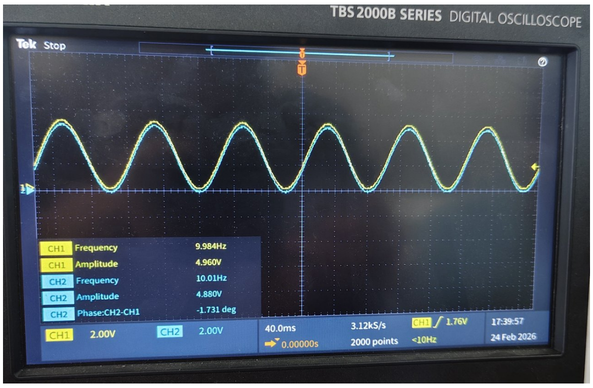
$$\implies \zeta = 1 \tag{109}$$

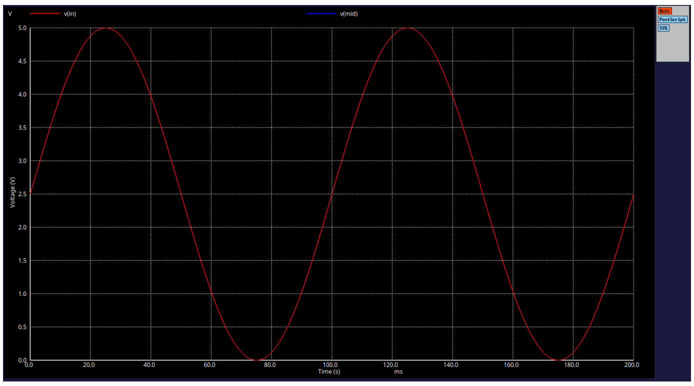
- As the value of $\zeta > \frac{1}{\sqrt{2}}$ ($\zeta = 1$) there will be no Resonance, band width, gain peaking in the case of critical damping

4.5.1 Wave Forms

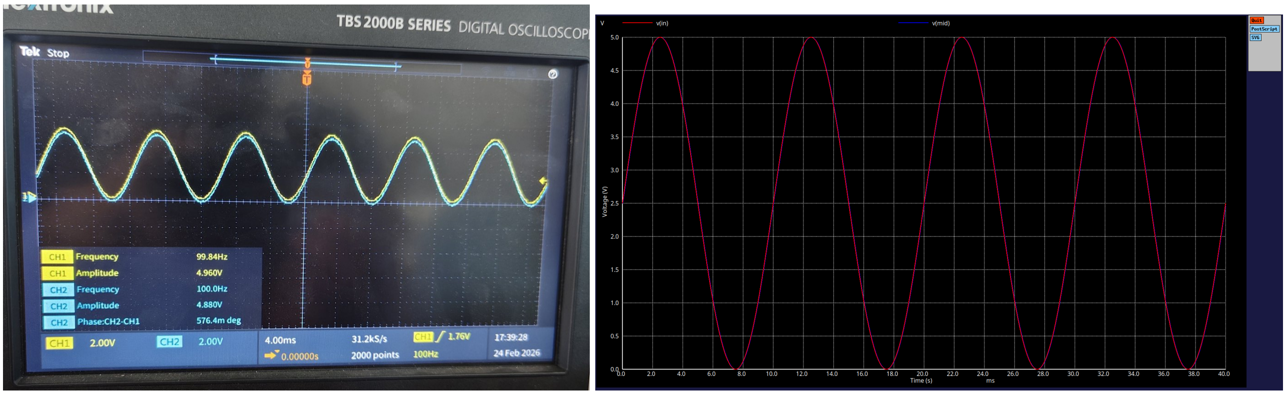


output waveform at 1HZ frequency

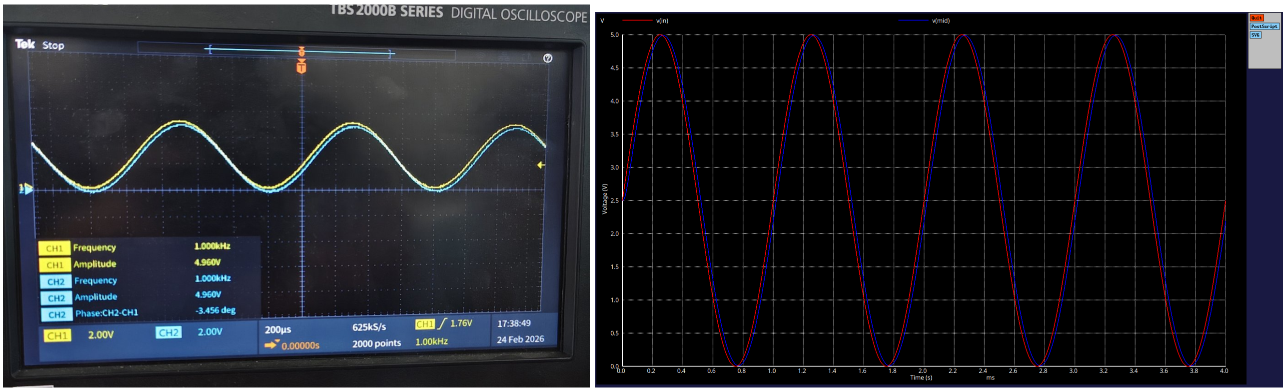




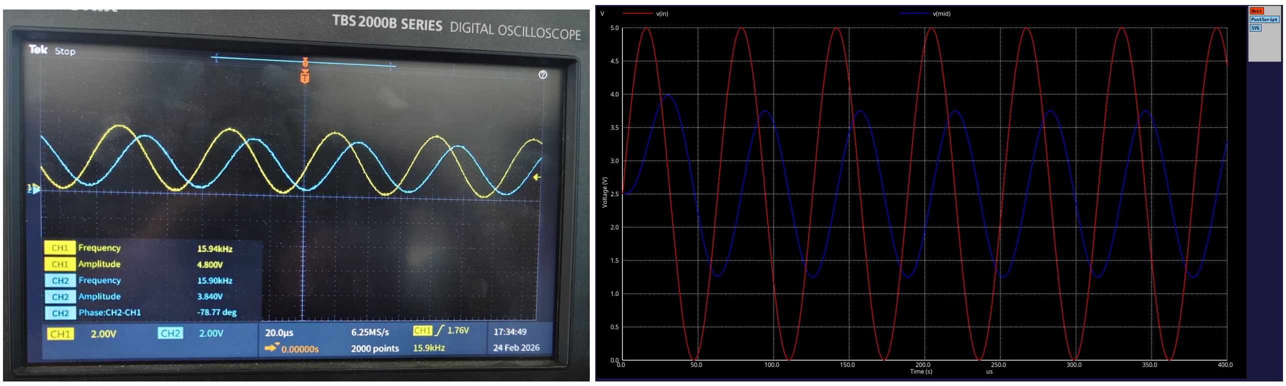
output waveform at 10HZ frequency



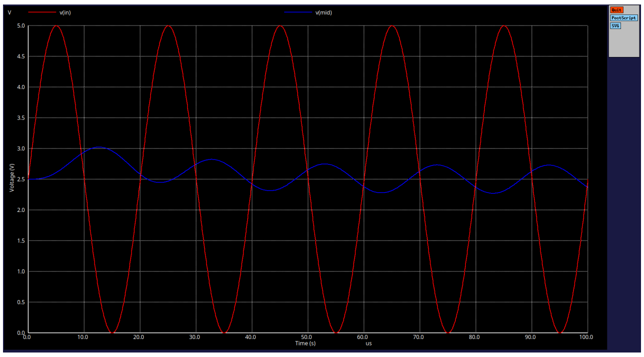
output waveform at 100HZ frequency



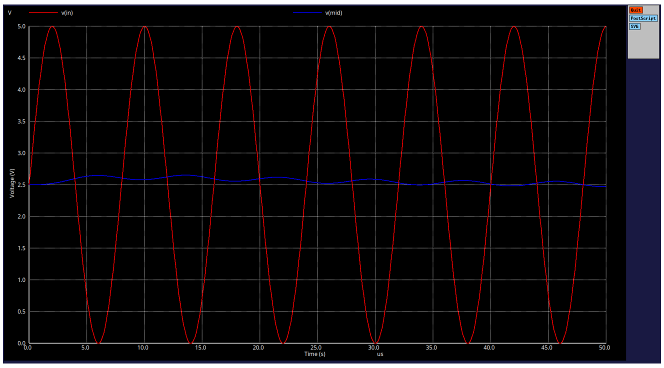
output waveform at 1KHZ frequency



output waveform at 15.9KHZ frequency

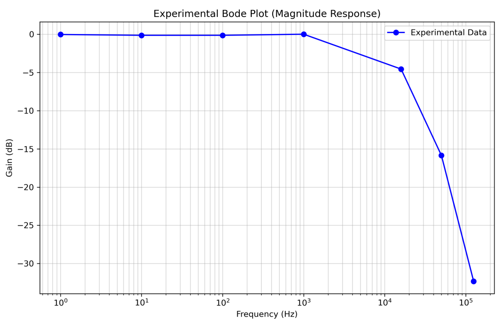


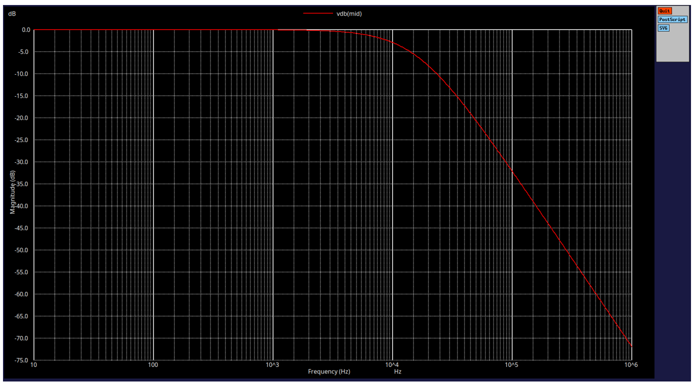
output waveform at 50KHZ frequency



output waveform at 100KHZ frequency

4.5.2 Bode plots





Plot with observed values and ngspice plot

Comparison table

Freq (Hz)	V_{in} (V)	V_{out} (V)	Gain (Exp)	Gain (Theo)	Error (%)
1	4.98	4.96	0.996	1.000	0.40
10	4.96	4.88	0.984	1.000	1.60
100	4.96	4.88	0.984	1.000	1.60
1,000	4.96	4.96	1.000	0.996	0.40
15,915	4.80	2.84	0.570	0.500	14.00
50,000	4.96	0.80	0.161	0.12	33.33
125,000	4.96	0.12	0.024	0.016	50.00

Comparison of Experimental and Theoretical Gain ($R = 2000\,\Omega$)

4.6 Under Damping

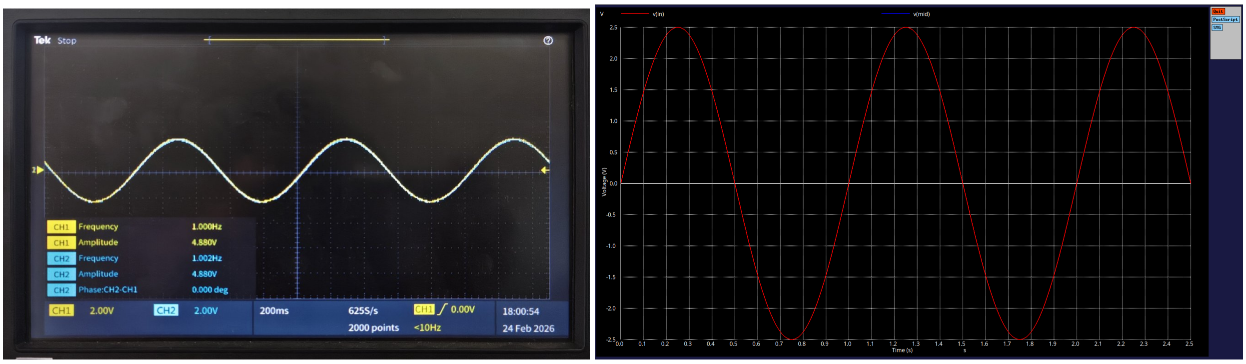
- For this case we have taken 220Ω resistance

$$\zeta = \frac{2200}{2000} \tag{110}$$

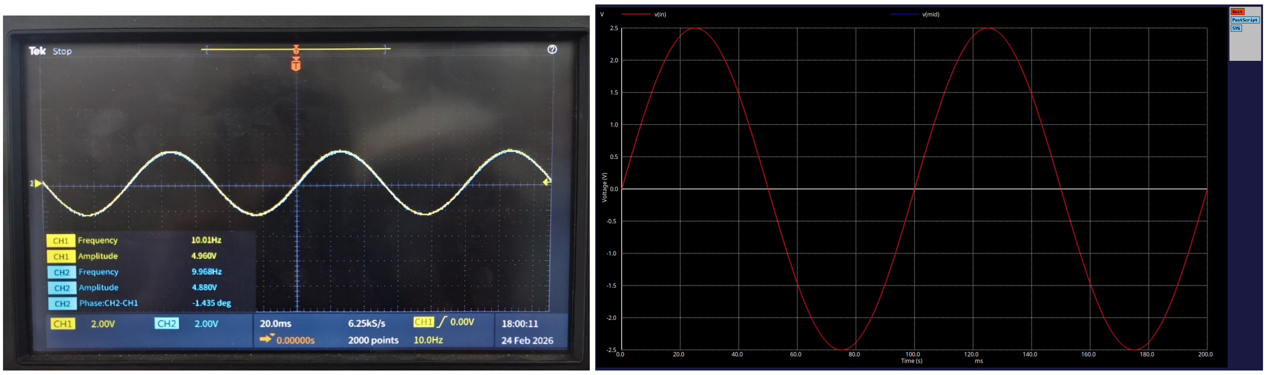
$$\implies \zeta = 0.11 \tag{111}$$

- The value of $\zeta < \frac{1}{\sqrt{2}}$ ($\zeta = 0.11$) there will be Resonance, band width, gain peaking in this case

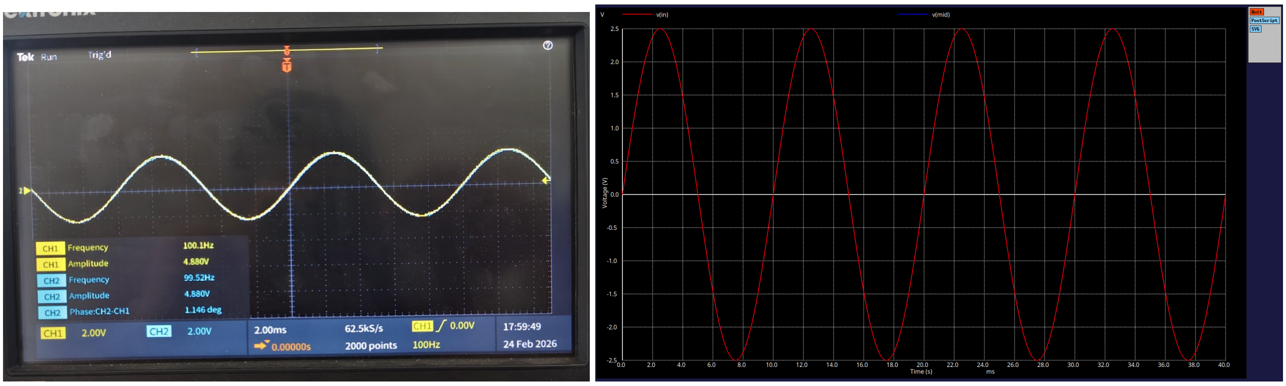
4.6.1 Wave Forms



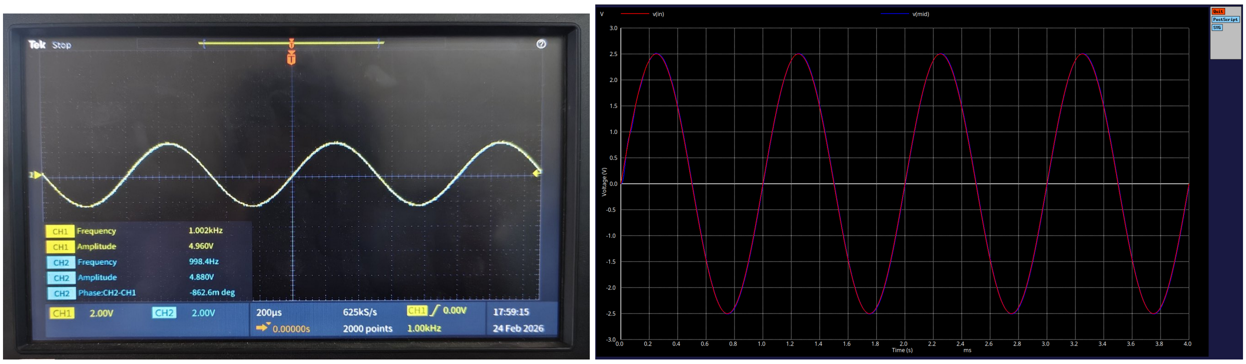
output waveform at 1Hz frequency



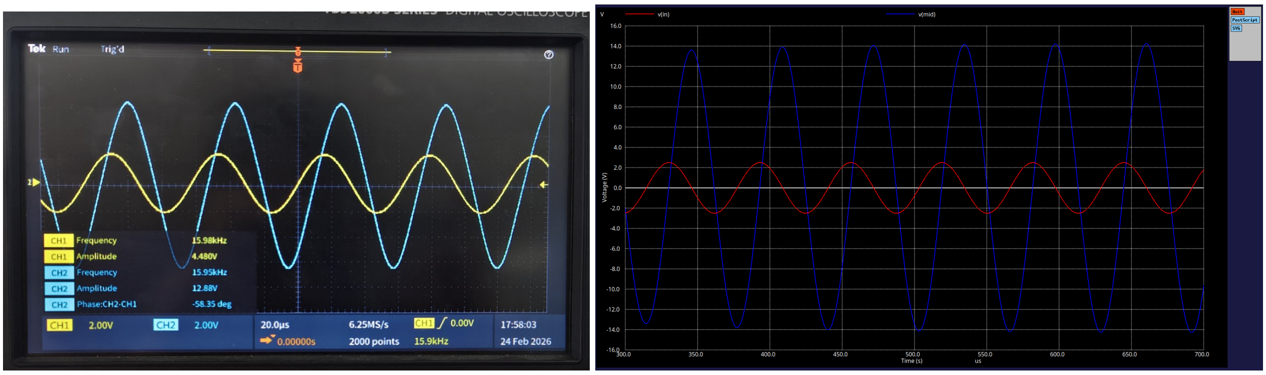
output waveform at 10Hz frequency



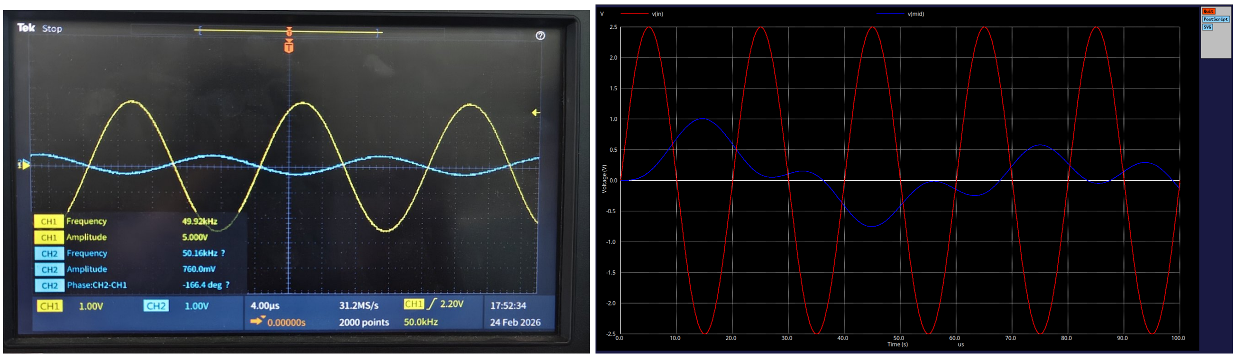
output waveform at 100Hz frequency



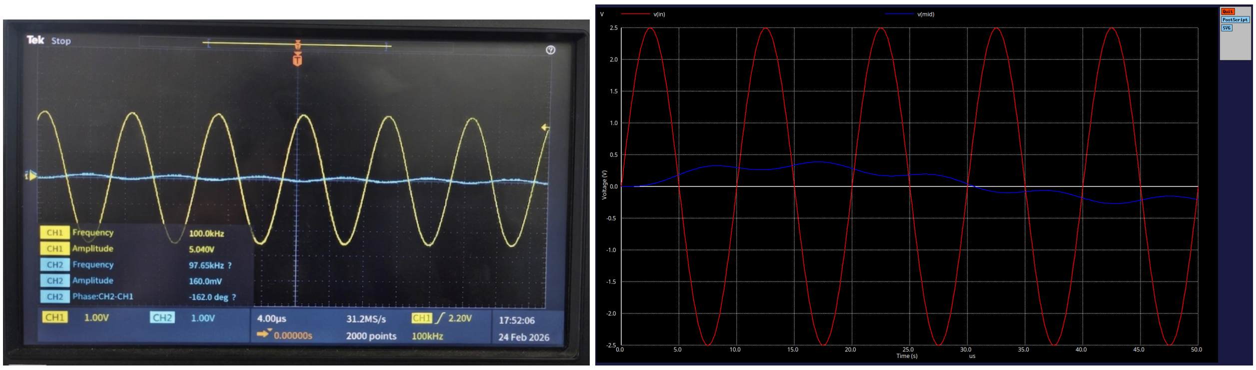
output waveform at 1KHz frequency



output waveform at 15.9KHz frequency



output waveform at 50KHz frequency



output waveform at 100KHz frequency

4.6.2 Resonant Frequency

$$f_r = f_n\sqrt{1 - 2\zeta^2} \tag{112}$$

- The value of ω_n is given by $\omega_n = \frac{1}{\sqrt{LC}} = 10^5$

- $f_n = \frac{10^5}{2\pi} \approx 15.92\,\text{KHz}$
- Simplifying the above expression we get $f_r \approx 15.72\,\text{KHz}$

4.6.3 Band Width

- $\text{BW} = \frac{R}{2\pi L} = \frac{22000}{2\pi}$
- Band width $\approx$ 3.5KHz

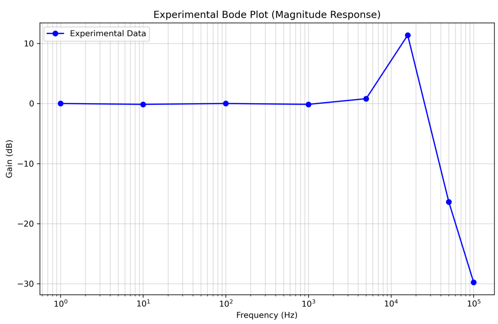
4.6.4 Gain Peaking

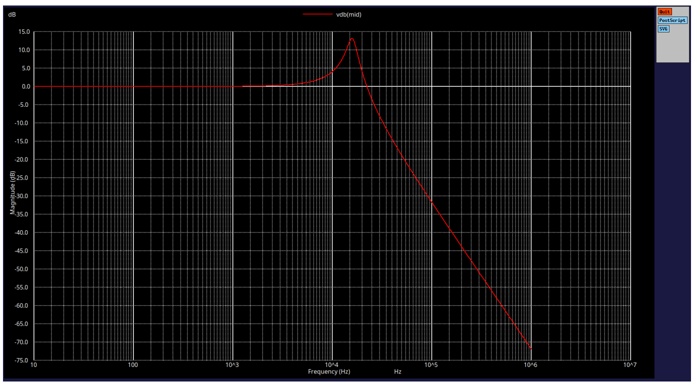
- Gain peaking in decibels is given by

$$\text{Gain Peaking (dB)} = 20\log_{10}\left(\frac{1}{2\zeta\sqrt{1-\zeta^2}}\right) \tag{113}$$

- Substituting the value of $\zeta = 0.11$ and on simplifying we get gain peaking = 13.2dB

4.6.5 Bode Plots





Plot with observed values and ngspice plot

Comparison table

Freq (Hz)	V_{in} (V)	V_{out} (V)	Gain (Exp)	Gain (Theo)	Error (%)
1	4.88	4.88	1.000	1.000	0.00
10	4.96	4.88	0.984	1.000	1.60
100	4.88	4.88	1.000	1.000	0.00
1,000	4.96	4.88	0.984	1.004	1.99
5,000	4.96	5.44	1.097	1.110	1.17
15,915	3.48	12.88	3.701	4.545	18.57
50,000	5.00	0.76	0.152	0.111	36.94
100,000	5.00	0.16	0.033	0.026	26.92

Comparison of Experimental and Theoretical Gain ($R = 220\,\Omega$)

4.7 Over Dampped

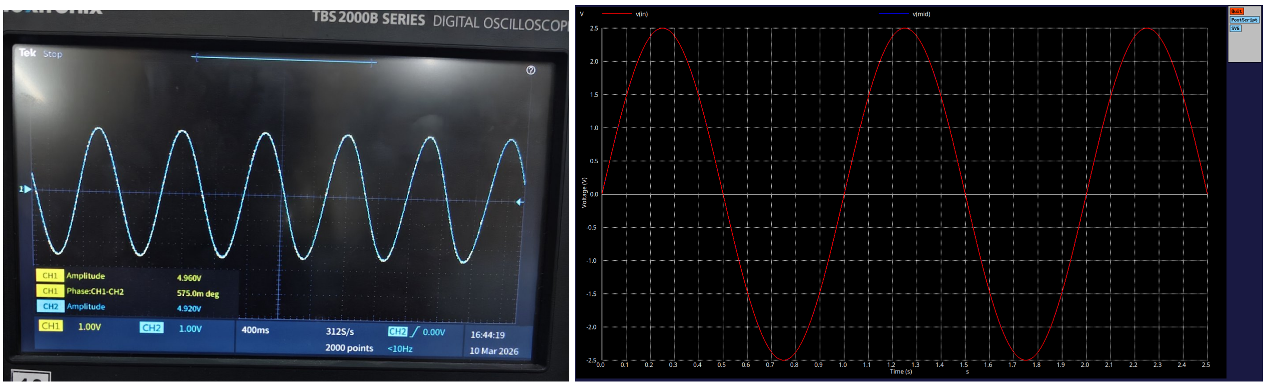
- For this case we have taken 10KΩ resistance

$$\zeta = \frac{10000}{2000} \tag{114}$$

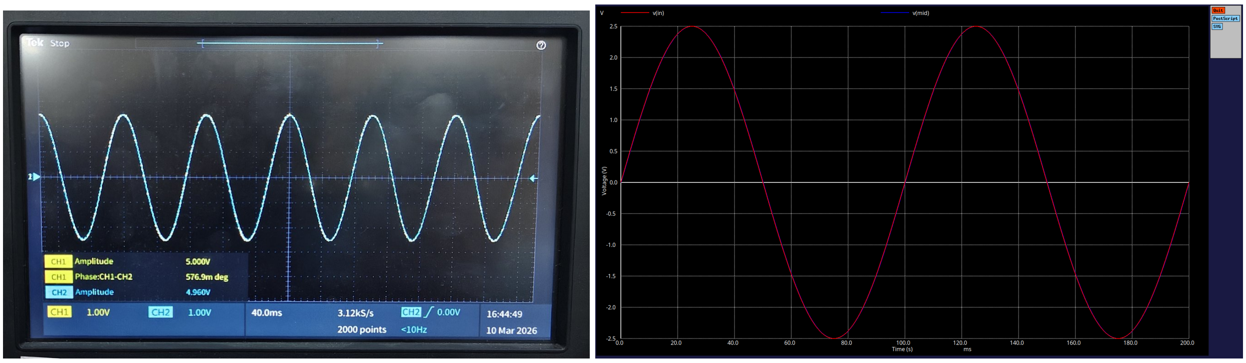
$$\implies \zeta = 5 \tag{115}$$

- As the value of $\zeta > \frac{1}{\sqrt{2}}$ ($\zeta = 5$) there will be no Resonance, band width, gain peaking in the case of critical damping

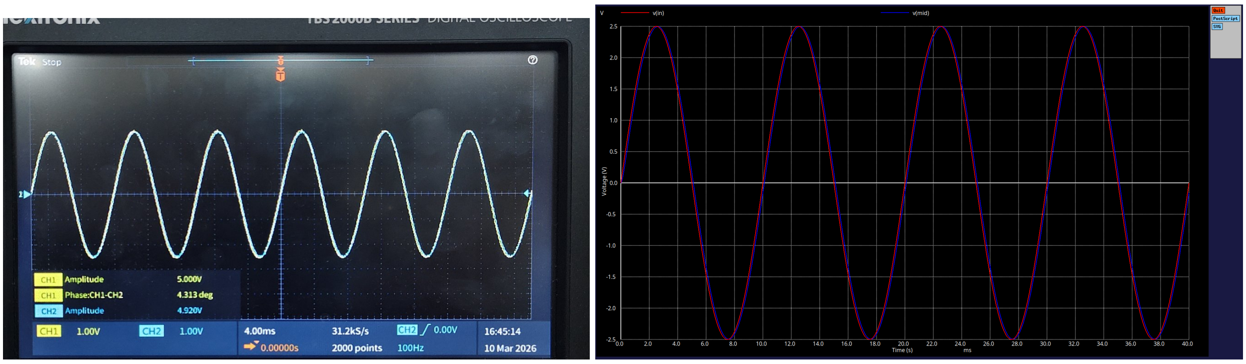
4.7.1 Wave forms



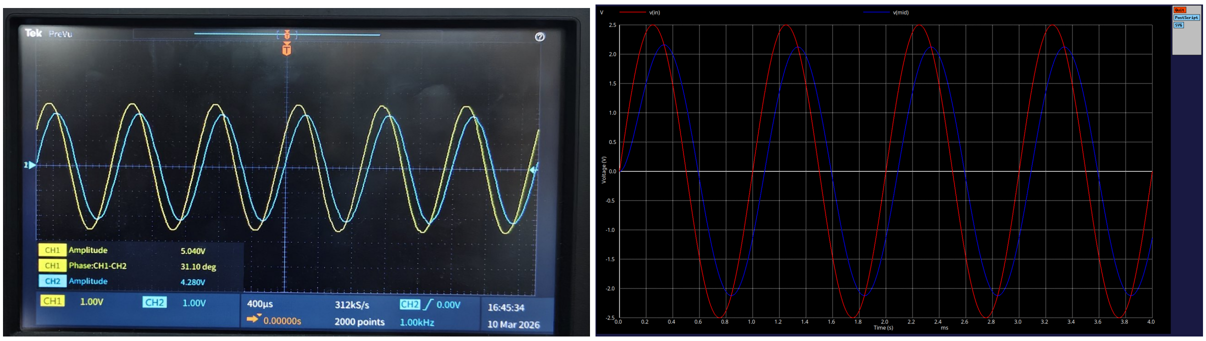
output waveform at 1Hz frequency



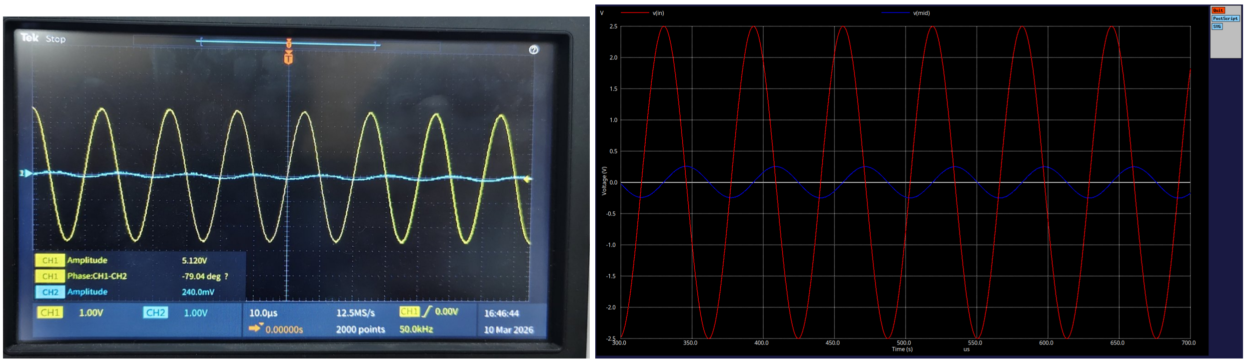
output waveform at 10Hz frequency



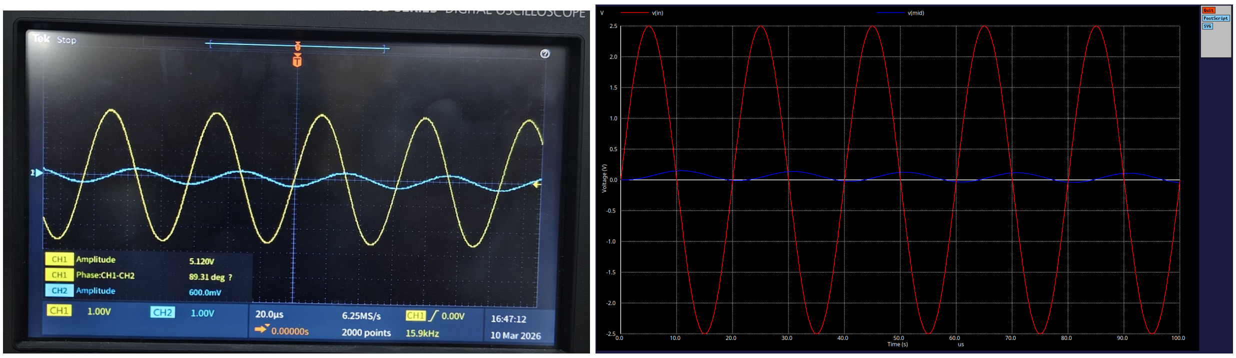
output waveform at 100Hz frequency



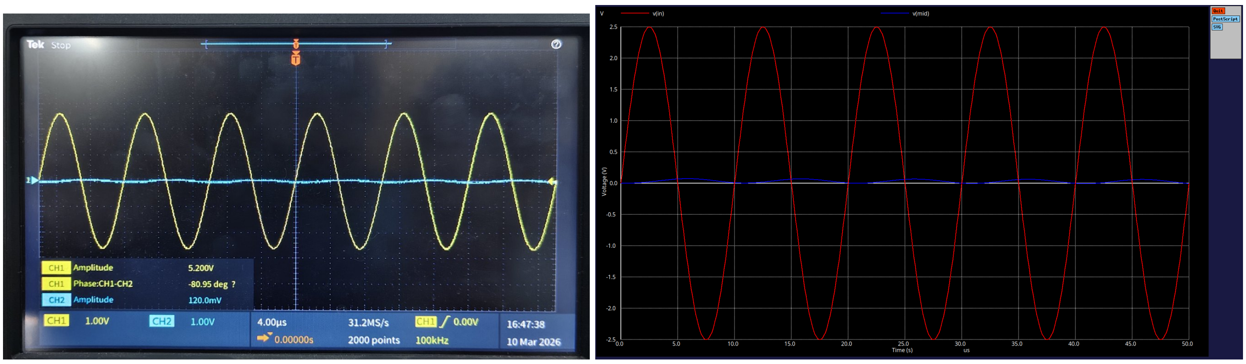
output waveform at 1KHz frequency



output waveform at 15.9KHz frequency

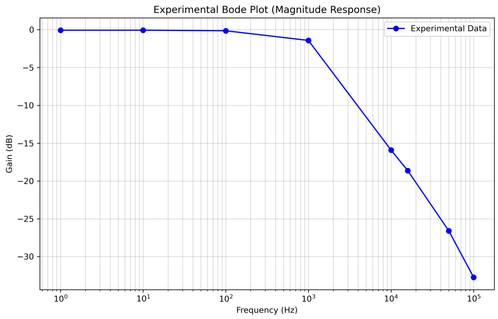


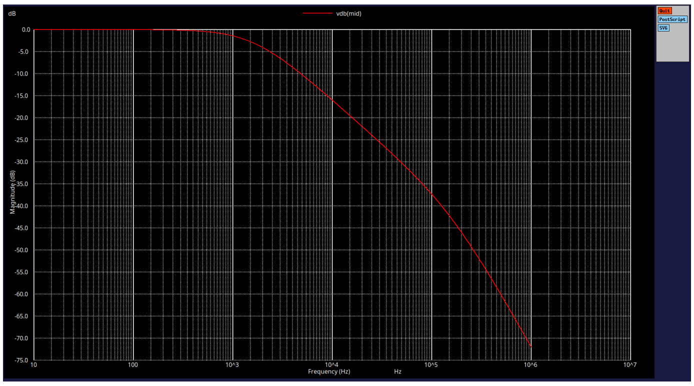
output waveform at 50KHz frequency



output waveform at 100KHz frequency

4.8 Bode Plots





Plot with observed values and ngspice plot

Comparison table

Freq (Hz)	V_{in} (V)	V_{out} (V)	Gain (Exp)	Gain (Theo)	Error (%)
1	4.96	4.92	0.992	1.000	0.81
10	5.00	4.96	0.992	1.000	0.80
100	5.00	4.92	0.984	0.998	1.41
1,000	5.04	4.28	0.849	0.849	0.01
10,000	5.12	0.92	0.180	0.158	13.42
15,915	5.12	0.60	0.117	0.100	17.18
50,000	5.12	0.24	0.047	0.039	20.5
100,000	5.20	0.12	0.023	0.017	35.3

Overdamped RLC Comparison ($R = 10,000\,\Omega$, $L = 10\,\text{mH}$, $C = 10\,\text{nF}$)

Effect of resistance on damping and bandwidth

For a second order series RLC Circuit the damping ratio(ζ) is given by

$$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}} \tag{116}$$

This shows that **damping ratio increases with the increase of resistance** when L,C are kept constant

Effect on band width

Band width is given by

$$BW = \frac{R}{2\pi L} \tag{117}$$

This shows that **Band Width increases with the increase of resistance** when L is kept constant

Conclusion :

For a series RLC circuit increase of resistance increases both the damping and band width

Relation between Time−Domain and Frequency response

For a second order series RLC system transfer function is given by

$$\implies H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}(\text{where } \omega_n = \frac{1}{\sqrt{LC}}) \tag{118}$$

Time−domain response

- small $\zeta < 1$ leads to oscillatory response with overshoot
- $\zeta = 1$ leads to non oscillatory response with less settling time
- $\zeta > 1$ leads to non oscillatory response with more settling time

Frequency-Domain response

- small $\zeta < 1$ leads to resonant condition, gain peaking in magnitude response
- $\zeta = 1$ leads to no peaking and smooth roll off and no resonance also
- $\zeta > 1$ leads to more gradual roll off compared to $\zeta = 1$ and no resonance also

Thus, the **damping constant(ζ) controls the both time and frequency responses**

5 Final analysis

5.1 Under damped ($\zeta < 1$)

- Oscillation of energy takes place between inductor and capacitor
- In time domain behavior it shows oscillations and results in overshoot

- In frequency domain it shows resonance and gain peaking($\zeta < \frac{1}{\sqrt{2}}$)
- So, Under damping case leads to resonance and peaking

5.2 Critical damping($\zeta = 1$)

- In time domain response it may show a very slight overshoot while settling and has less settling time than over damped
- In frequency response no resonance and no peaking occurs
- So, Critical damping represents the boundary condition

5.3 Over damping($\zeta > 1$)

- High resistance leads to high value of damping constant with no oscillations in time domain response
- Has more settling time than critical damping
- In frequency response no resonance and no peaking occurs
- So, Overdamped response corresponds to no resonance